

From hyperspectral CHRIS to four RGBN bands

The cloud model eats **4 bands** (R, G, B, NIR). CHRIS has **18 to 62 bands** per mode. So each pseudo-band is built from the CHRIS bands that fall in it.

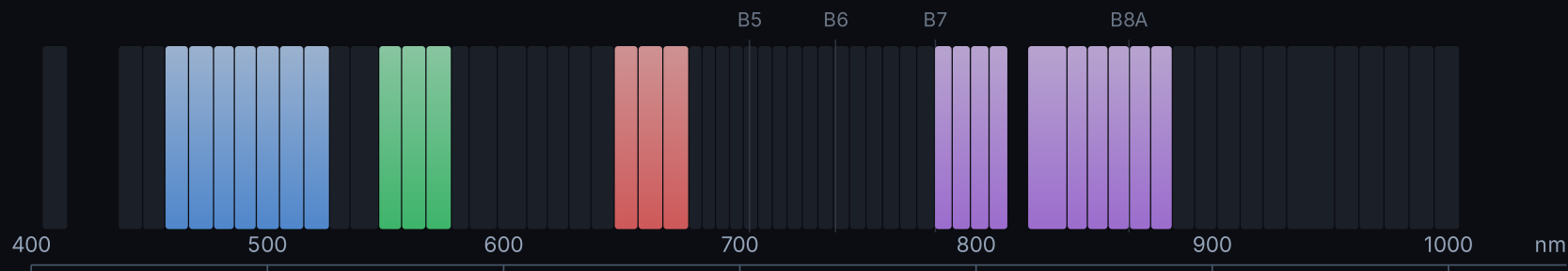
which each CHRIS band goes to the Sentinel-2 band it overlaps the most; in the NIR the model also averaged the red-edge band beside it

how plain **mean** of the CHRIS bands assigned to each pseudo-band

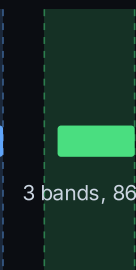
then DN/100 000 or TOA/10 000, clip to 5, and a per-tile z-score inside the network

Training mapping, the one the model actually learned. The HuggingFace loader will be aligned to this.

Mode 1 : 62 bands, ~34 m



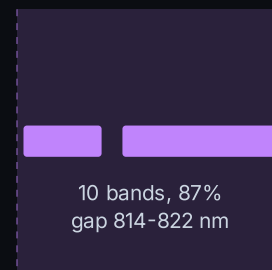
B2, 459-525 nm



B3, 542-578 nm



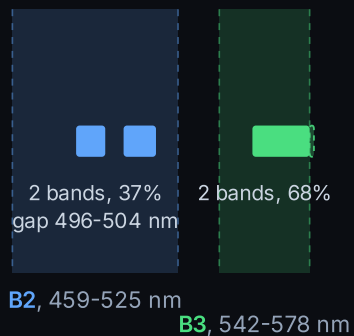
B4, 649-680 nm



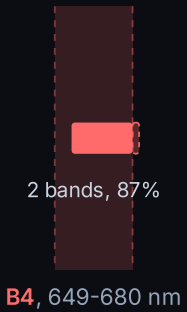
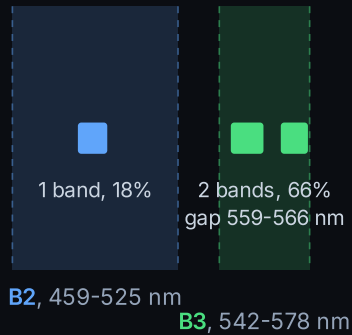
B8, 780-886 nm

NIR takes 42-51: band 42 also falls fully inside B8 (it tied with B7) and the model averaged it in.

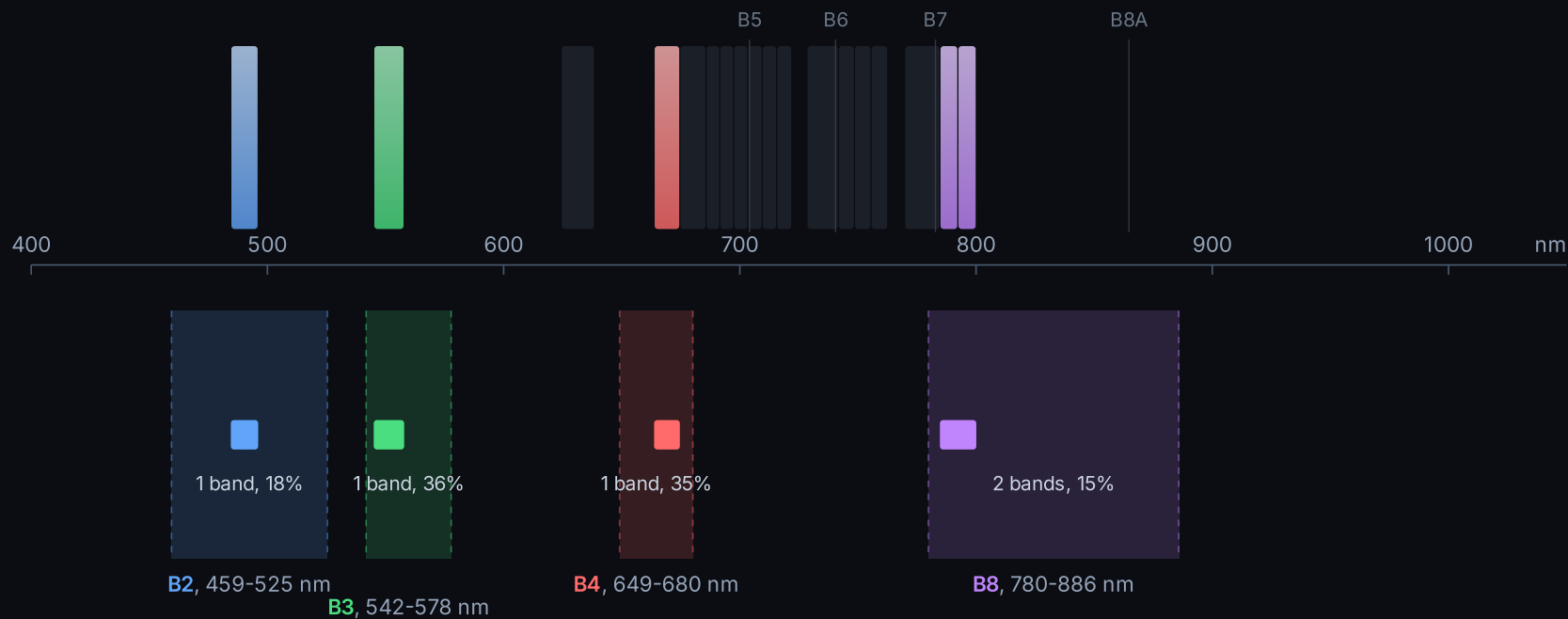
Mode 2 : 18 bands, ~17 m



Mode 3 : 18 bands, ~13 m

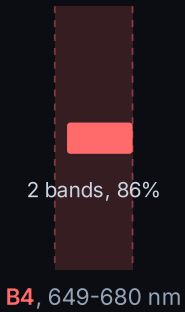
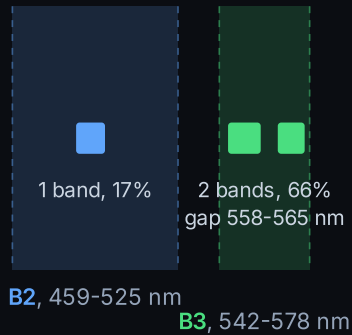
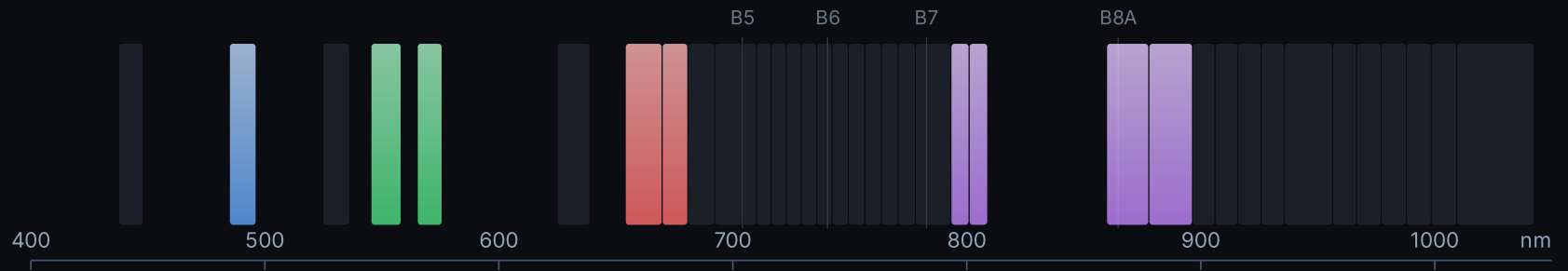


Mode 4 : 18 bands, ~11 m

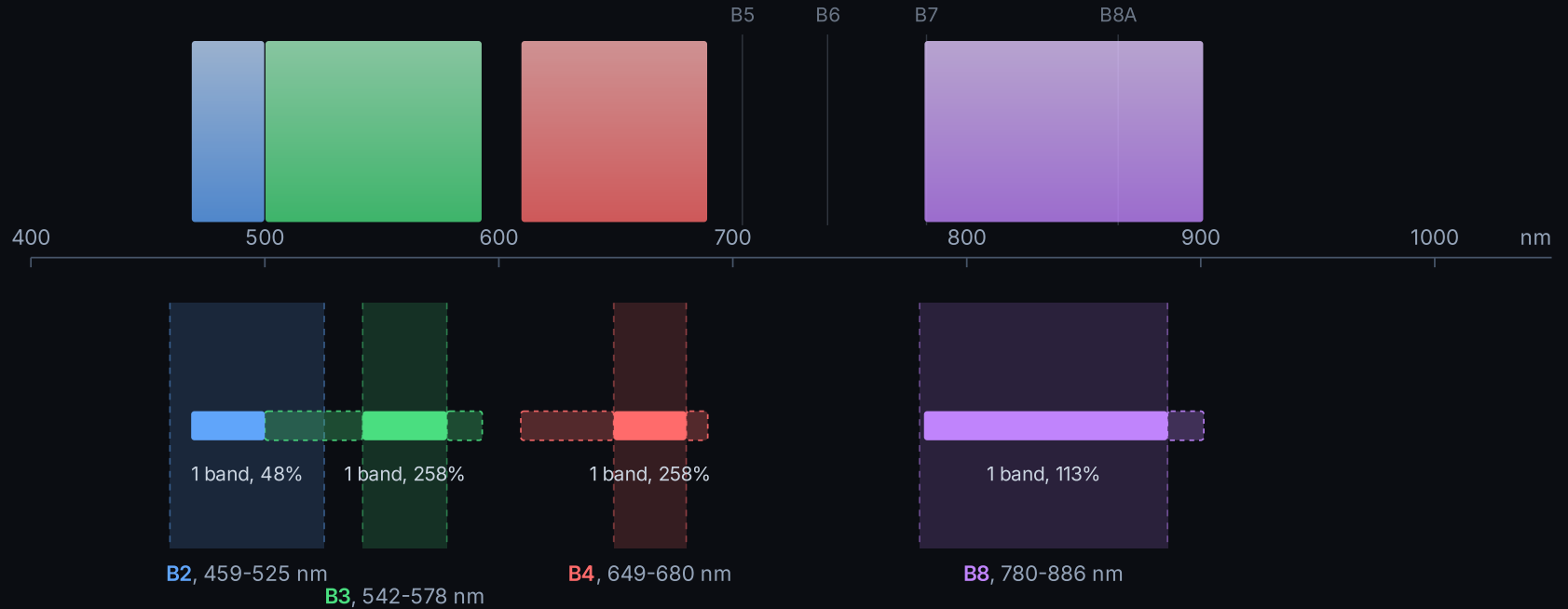


Mode 4 has one band per target, and NIR also takes the red-edge band 17: it overlaps B8 fully.

Mode 5 : 37 bands, ~20 m



Mode 20 : 4 bands, ~17 m, DN only



Mode 20 has exactly 4 bands: no averaging. Each one is wide and sits shifted from the S2 range it falls into.

Where the bands end up

R, B4

G, B3

B, B2

NIR, B8

Mode	B2 blue	B3 green	B4 red	B8 NIR
1	4-10 (7), 457-526 nm	13-15 (3), 547-578	23-25 (3), 647-678	42-51 (10), 782-883
2	3-4 (2), 485-517	6-7 (2), 555-580	10-12 (3), 644-686	16-17 (2), 771-887
3	2 (1), 486-497	4-5 (2), 546-577	7-8 (2), 656-683	14-15 (2), 773-890
4	1 (1), 484-496	2 (1), 545-558	4 (1), 664-675	17-18 (2), 785-800
5	2 (1), 485-496	4-5 (2), 545-576	7-8 (2), 654-681	23-26 (4), 793-896
20	1	2	3	4

Training mapping. Numbers are 1-based CHRIS band indices; nm is the spectral span. Stack order: R, G, B, NIR.